

Data and Visualisations

Social and Cultural Analytics (7AAVBCS5)

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Lecture Plan

- **Part I: Introduction to Data Visualisation**
- **Part II: Bar plots for categorical data**
- **Part III: Distributions & Central Tendency**
- **Part IV: Histograms for numeric data**
- **Part V: Conclusion**
- **Part VI: Live Coding**

Key Learning Outcomes

By the end of this week you will be able to:

- Choose a visualisation appropriate to your data type
- Build and customise bar plots, including stacked, grouped and proportional
- Describe a distribution and read its shape

- Build and customise histograms
- Save figures to file at publication quality
- Recognise a misleading chart

Mid-term feedback

How is it going?

(One good) What is one thing that is working well for you in this module?

(One improvement) What is one thing that could be improved?

My one request for you all!

Review: what we've learned so far

- Types, variables, lists, arrays, dictionaries
- Comparisons and `if/elif/else`
- Data frames: building, inspecting, selecting, filtering, `.loc/.iloc`
- Reading and saving CSV
- Missing data: quantifying it, and asking whether it is patterned
- Categories, frequency tables, cross-tabulation, `groupby`
- **Counts versus percentages**, and which margin answers which question

This week: turning those tables into pictures

Part I: Introduction to Data Visualisation

Exploratory Data Analysis

You (researcher, analyst, historian) sit between **the data** and **the reader**.

Visualisation does two different jobs:

- **For you** — exploratory. What shape is this? What is odd about it? What should I look at next?
- **For the reader** — explanatory. Here is the thing I found, and here is why you should believe it.

Most of the plots you make this term are for the first purpose, and you will throw them away.

Visualisations are not always best

1 7 1 8 9 0 4 6 2 8 4 3 8 5 1 8 6 2 7 5 4 2 3

Sometimes a number, a sentence, or a small table communicates better than a chart.

Why visualise, then?

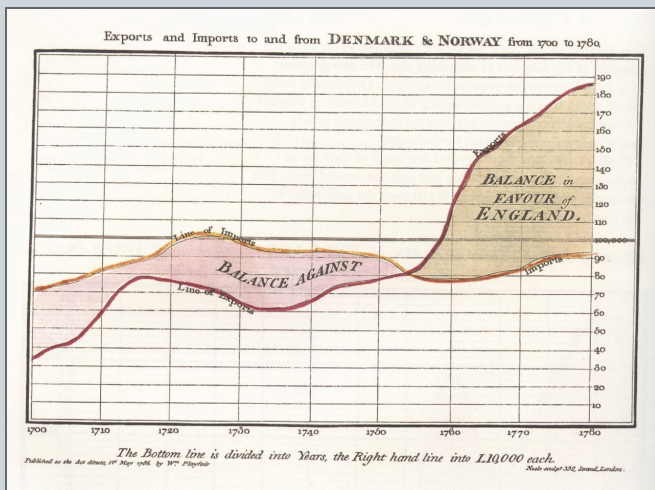
- **Pattern recognition** — human brains process visual information faster than text or tables
- Reveals outliers and gaps that summary statistics hide
- Shows *shape*, which no single number can
- Communicates to readers who will not read your table

Visualisation is a field of research in itself

William Playfair (1759–1823), *The Commercial and Political Atlas* — the invention of the line chart, bar chart and pie chart.

Charles Joseph Minard (1781–1870) — the Russian campaign of 1812, six variables in one figure.

Maps as visualisations that also tell stories.



Choosing the right visualisation

Data type	Example variables	Best visualisation
Categorical (nominal)	country, genre, gender	Bar plots
Categorical (ordinal)	rating, education level	Bar plots (ordered), stacked bars
Numeric (discrete)	counts, years	Bar plots for counts, line plots for time
Numeric (continuous)	age, temperature, price	Histogram, density, box plots
Two categorical	region × political affiliation	Grouped/stacked bars, heatmaps
Numeric and categorical	age × country	Box plots, violin plots, scatter

Part II: Bar plots for categorical data

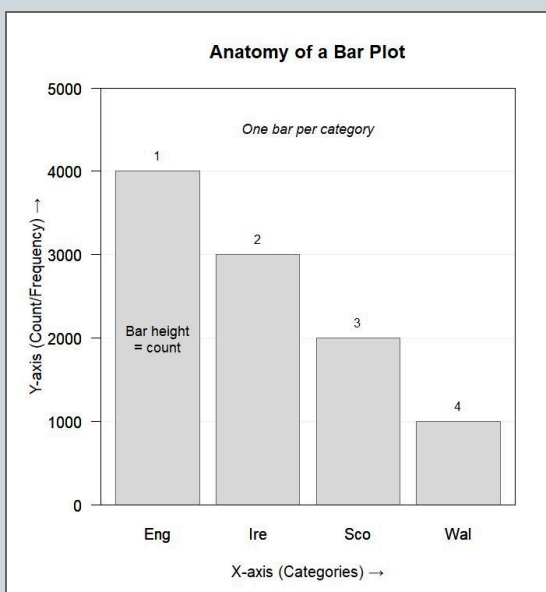
Anatomy of a bar plot

What bar plots show

- Frequencies or counts of a categorical variable
- Height of bar = count for that category
- One bar per category

When to use

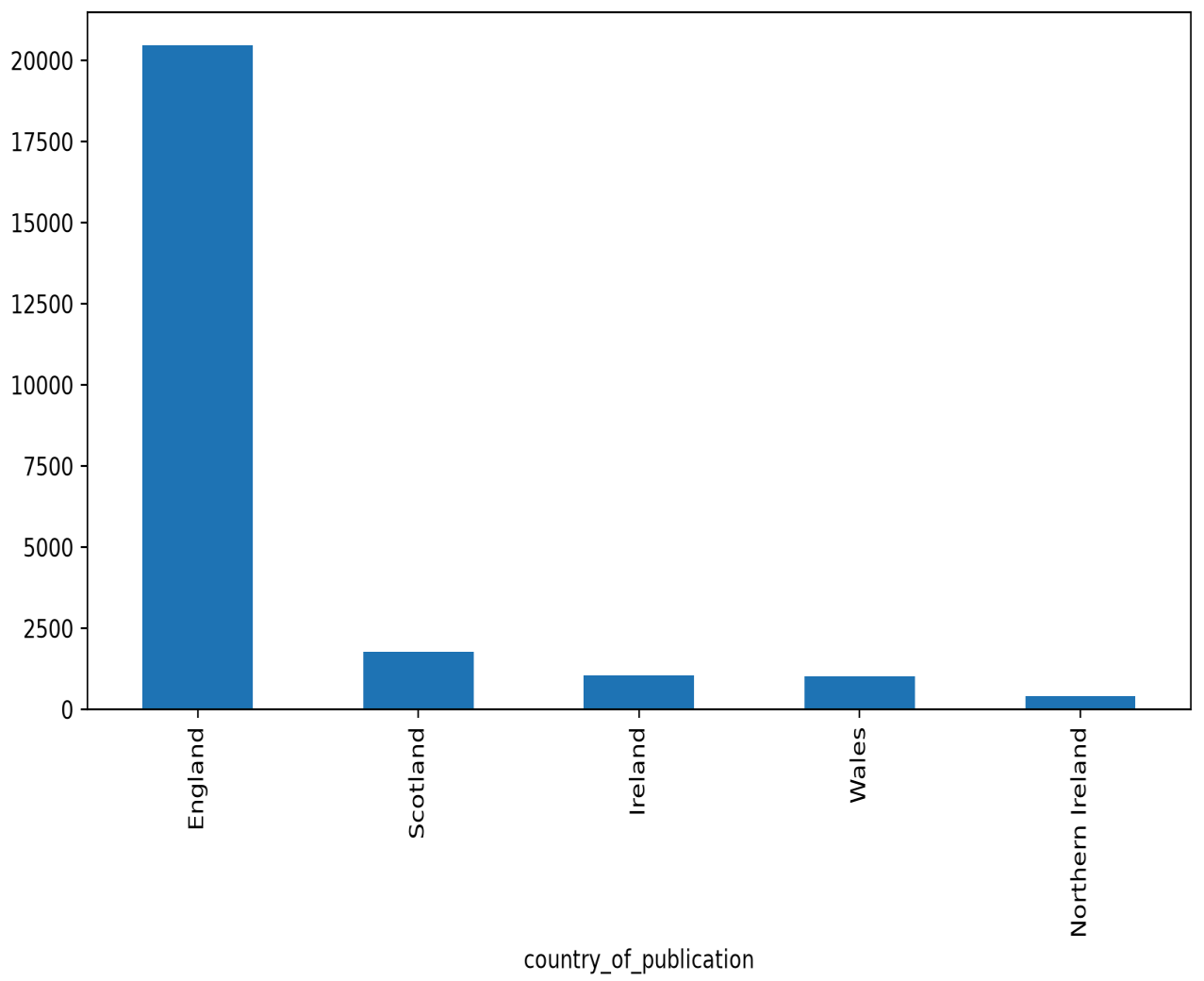
- Comparing counts across categories
- Showing the composition of a dataset
- Displaying survey responses



Creating a bar plot

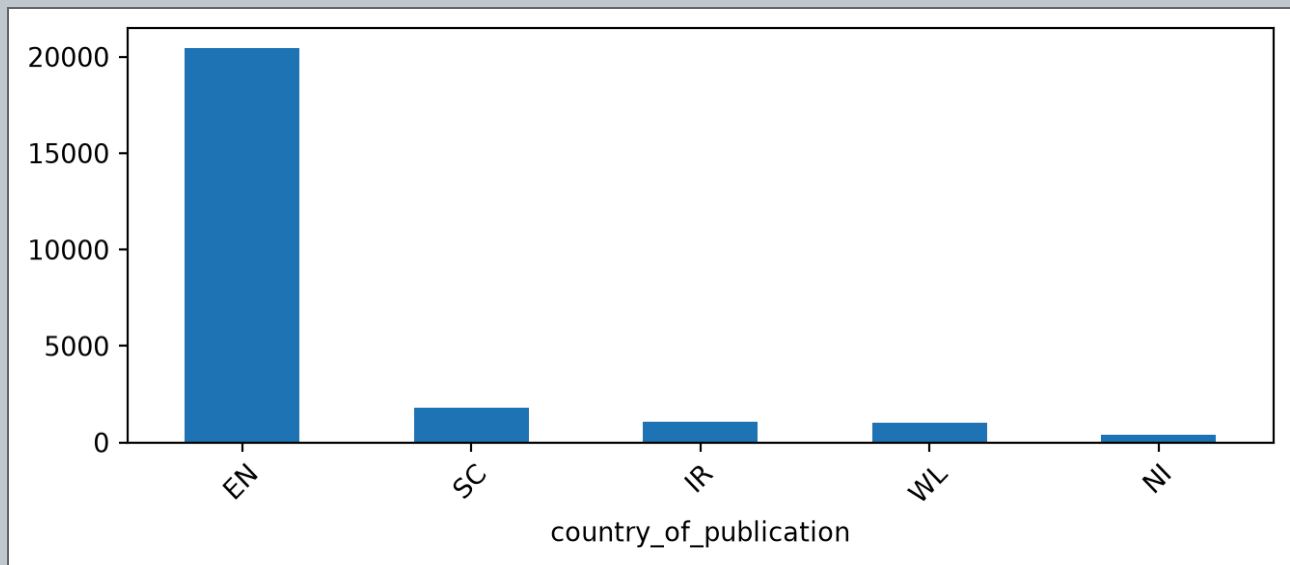
```
1 import matplotlib.pyplot as plt
2 import pandas as pd
3
4 news = pd.read_csv("../data/BritishAndIrishNewspapersTitleList_20191118.csv",
5                     low_memory=False)
6 countries_top = news["country_of_publication"].value_counts().head(5)
7 print(countries_top)
8
9 countries_top.plot(kind="bar")
10 plt.show()
```

```
country_of_publication
England                20465
Scotland               1778
Ireland                1050
Wales                  1019
Northern Ireland        415
Name: count, dtype: int64
```



Fixing the labels

```
1 fig, ax = plt.subplots(figsize=(7, 3))
2 countries_top.plot(kind="bar", ax=ax, rot=45)
3 ax.set_xticklabels(["EN", "SC", "IR", "WL", "NI"])
4 plt.tight_layout()
5 plt.show()
```



R

Python

`las = 2`

`rot=90 (or any angle)`

`cex.names = 0.5`

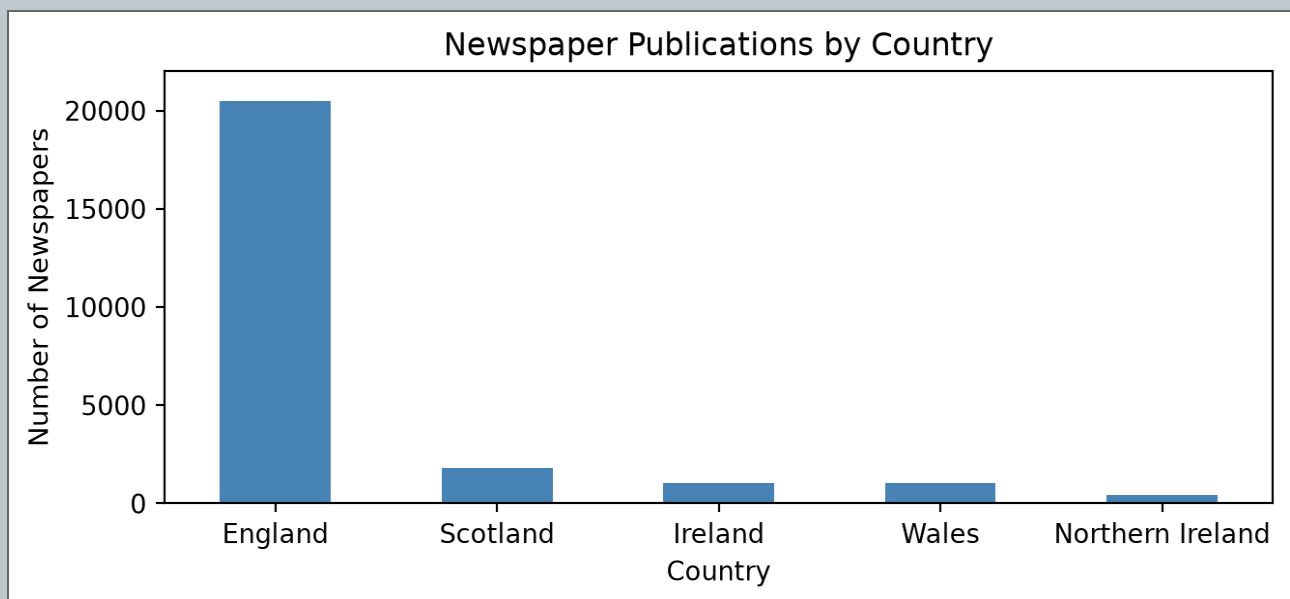
`ax.tick_params(labelsize=8)`

`names.arg = c(...)`

`ax.set_xticklabels([...])`

Titles, colours, limits

```
1 fig, ax = plt.subplots(figsize=(7, 3.2))
2 countries_top.plot(kind="bar", ax=ax, color="steelblue", rot=0)
3 ax.set_title("Newspaper Publications by Country")
4 ax.set_xlabel("Country")
5 ax.set_ylabel("Number of Newspapers")
6 ax.set_ylim(0, 22000)
7 plt.tight_layout()
8 plt.show()
```



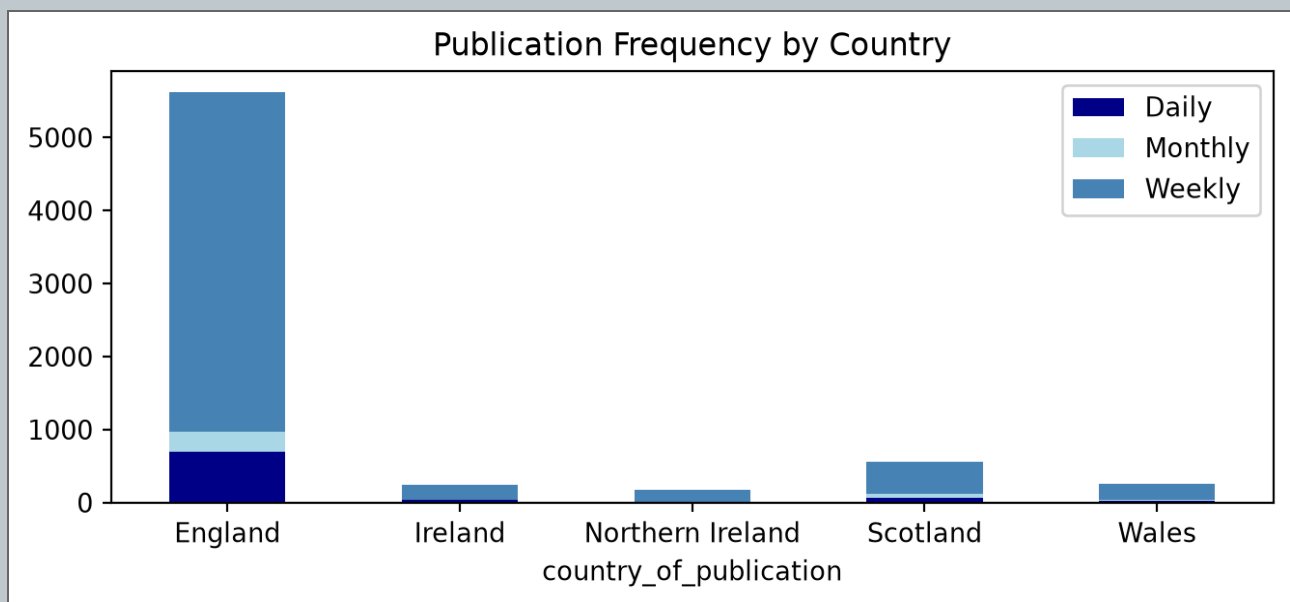
From crosstab to chart

```
1 top5 = ["England", "Scotland", "Ireland", "Wales", "Northern Ireland"]
2 subset = news[news["country_of_publication"].isin(top5)
3           & news["current_publication_frequency"].isin(["Weekly", "Daily", "Monthly"])]
4
5 countries_pf = pd.crosstab(subset["current_publication_frequency"],
6                           subset["country_of_publication"])
7 countries_pf
```

country_of_publication	England	Ireland	Northern Ireland	Scotland	Wales
current_publication_frequency					
Daily	698	33	10	59	28
Monthly	271	6	3	55	4
Weekly	4654	208	161	442	228

Stacked bars

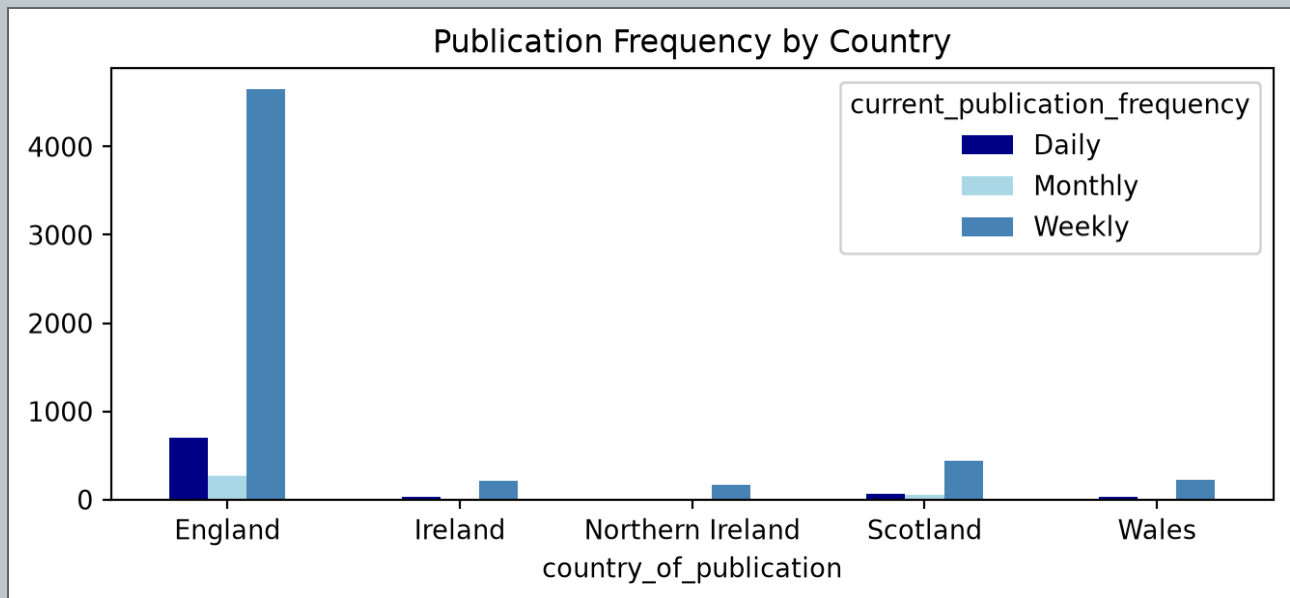
```
1 fig, ax = plt.subplots(figsize=(7, 3.2))
2 countries_pf.T.plot(kind="bar", stacked=True, ax=ax, rot=0,
3                       color=["darkblue", "lightblue", "steelblue"])
4 ax.set_title("Publication Frequency by Country")
5 ax.legend(title="")
6 plt.tight_layout()
7 plt.show()
```



- Total height = total titles per country
- Segment height = count for each frequency
- Good for **composition and totals at once**

Grouped bars

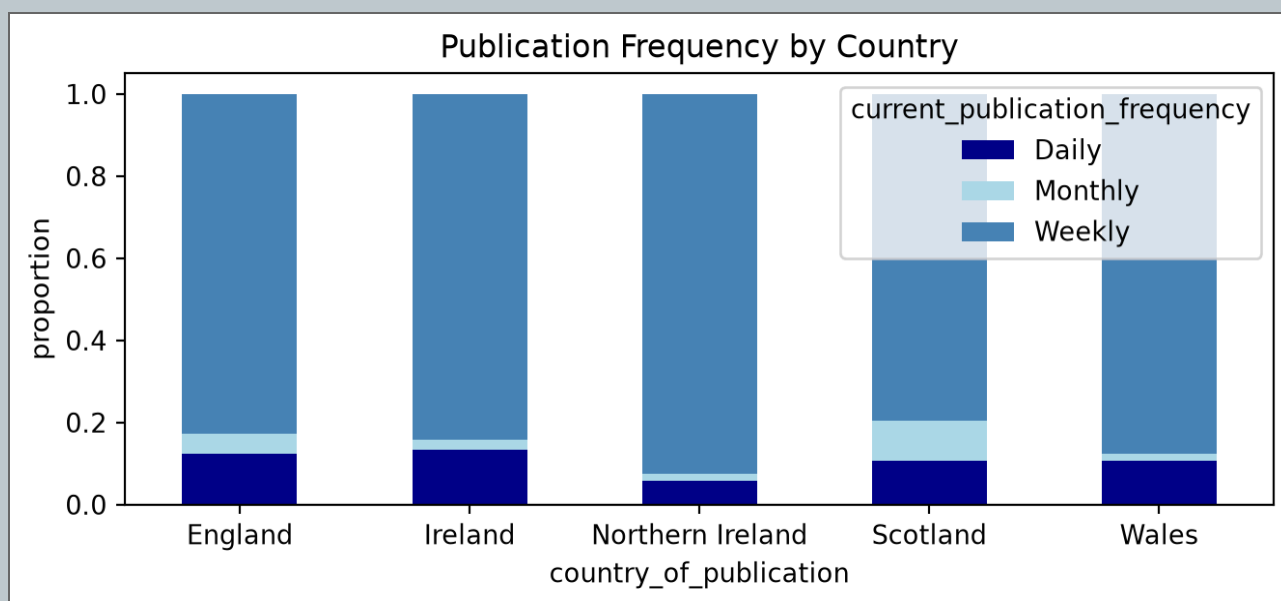
```
1 fig, ax = plt.subplots(figsize=(7, 3.2))
2 countries_pf.T.plot(kind="bar", ax=ax, rot=0,
3                       color=["darkblue", "lightblue", "steelblue"])
4 ax.set_title("Publication Frequency by Country")
5 plt.tight_layout()
6 plt.show()
```



- Every bar starts from zero, so values are directly comparable
- Good for **quickly seeing differences**

Proportional stacked bars

```
1 props = pd.crosstab(subset["current_publication_frequency"],
2                     subset["country_of_publication"], normalize="columns")
3
4 fig, ax = plt.subplots(figsize=(7, 3.2))
5 props.T.plot(kind="bar", stacked=True, ax=ax, rot=0,
6             color=["darkblue", "lightblue", "steelblue"])
7 ax.set_ylabel("proportion")
8 ax.set_title("Publication Frequency by Country")
9 plt.tight_layout()
10 plt.show()
```



Stacked vs grouped vs proportional

	Stacked (count)	Proportional	Grouped (count)
Shows	totals clearly	proportions clearly	individual values clearly
Proportions	okay	great for comparing across categories, bad for raw counts	good for comparisons
Space	efficient	either	needs more
Weakness	hard to compare middle segments, or categories with very different totals	hard to compare middle segments	—

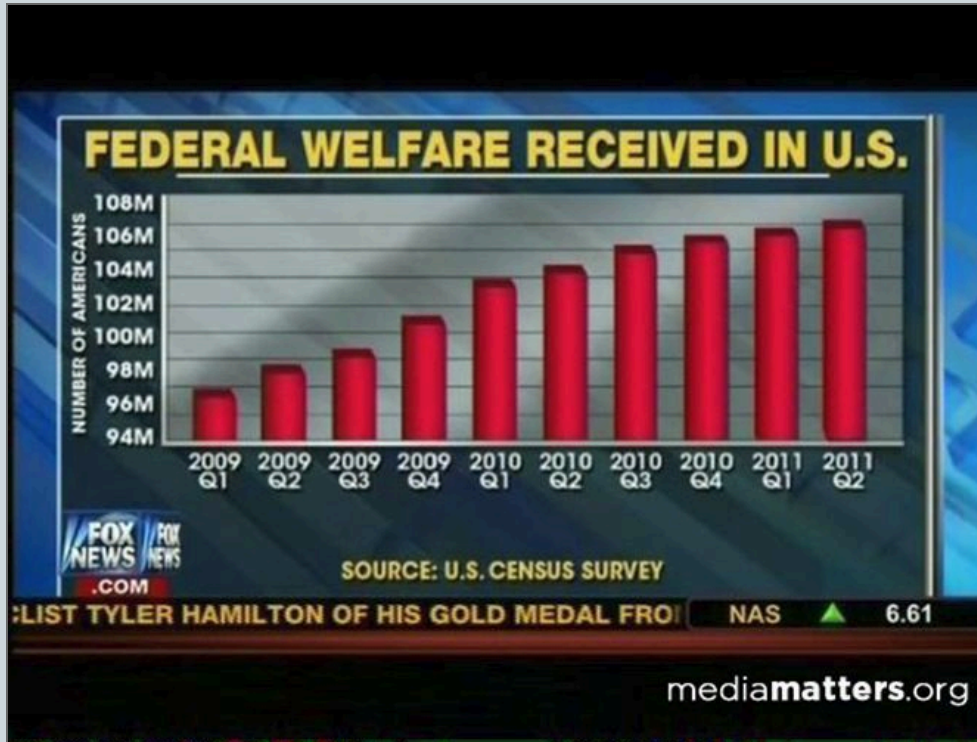
Saving figures

```
1 from pathlib import Path
2 Path("figures").mkdir(exist_ok=True)
3
4 fig, ax = plt.subplots(figsize=(8, 6))
5 countries_pf.T.plot(kind="bar", stacked=True, ax=ax)
6 fig.savefig("figures/barplot_countries.png", dpi=150, bbox_inches="tight")
7 fig.savefig("figures/barplot_countries.pdf", bbox_inches="tight")
```

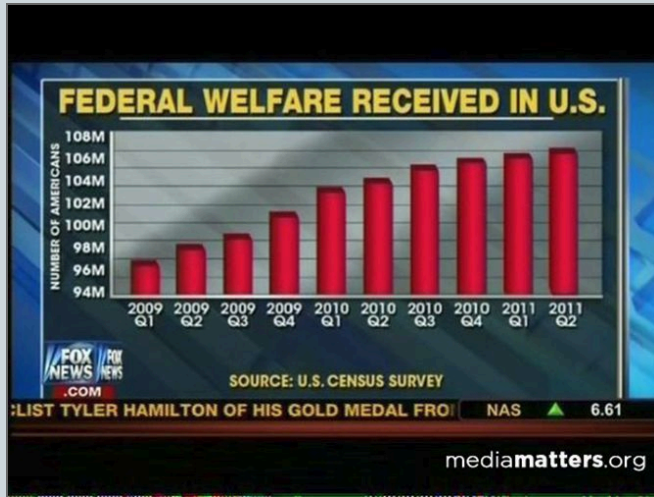
- **PNG** for web and presentations; **PDF** for publication
- `bbox_inches="tight"` stops labels being cut off
- No `dev.off()` needed

Bad visualisations

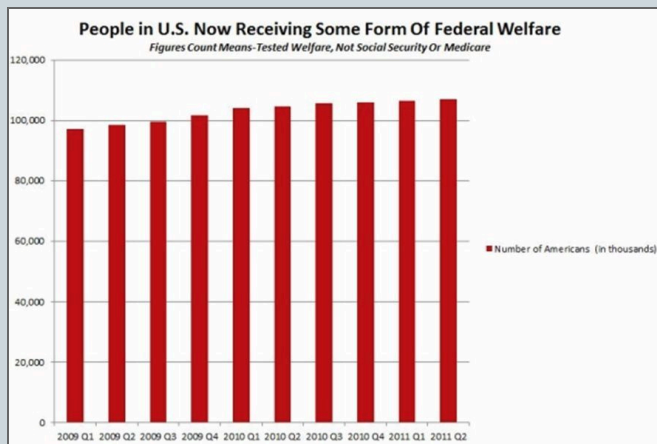
Bad charts are often bad on purpose



The truncated axis

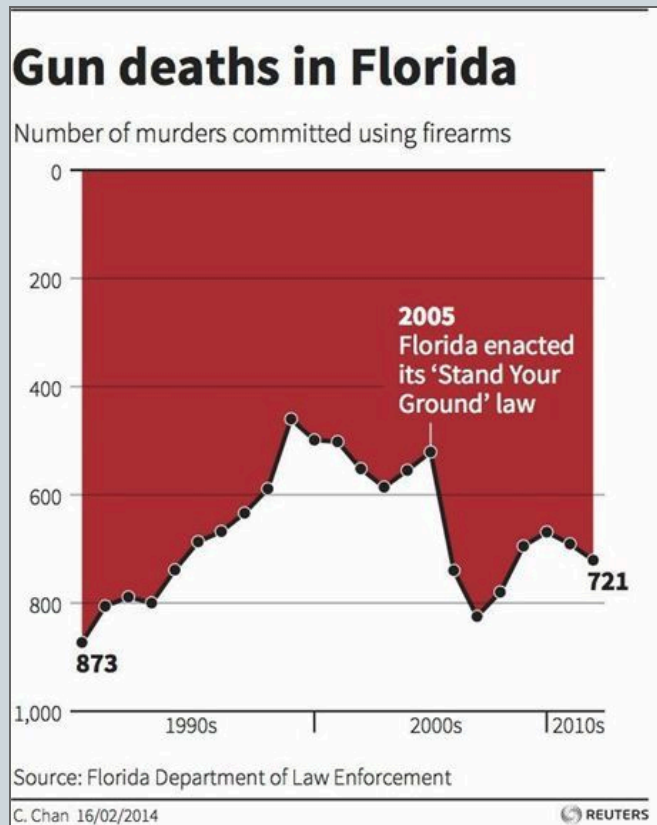


The y-axis has been purposefully truncated to make the change look larger than it is.



Same data, complete y-axis.
A very different story emerges.

Flipping the axis

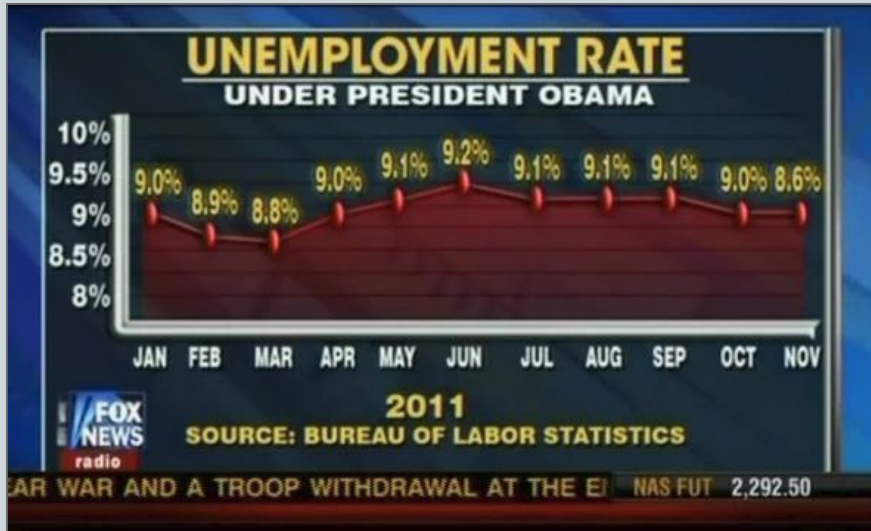


Effect of Florida's "stand your ground" law (2005), as published by the Florida Department of Law Enforcement.

At a glance: gun deaths have dropped since the law came in.

The y-axis is flipped.
Deaths have increased.

And sometimes they are simply wrong



Part III: Distributions & Central Tendency

Mean, median, mode

- **Mean:** add up all the numbers, divide by how many there are
- **Median:** the middle number, once ordered
- **Mode:** the most commonly occurring value

Central tendency describes how observations bunch around a value.

Why distributions matter

- Report the **shape**:
 - Symmetric (the “bell”)
 - Skewed (a long tail)
 - Bimodal (two or more peaks)
- Reveal outliers
- Indicate appropriate statistical methods

- Tell stories about phenomena

If we have averages, why visualise?

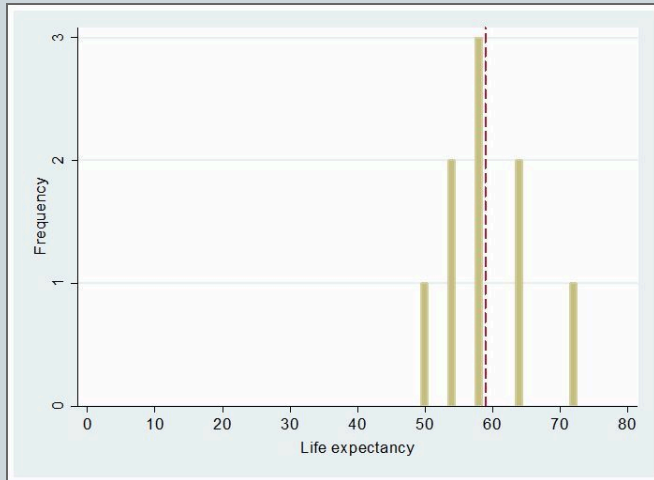
Because the average of a distribution can be nothing like a typical case.

Life expectancy in Britain 400 years ago was 39. Today it is 81.

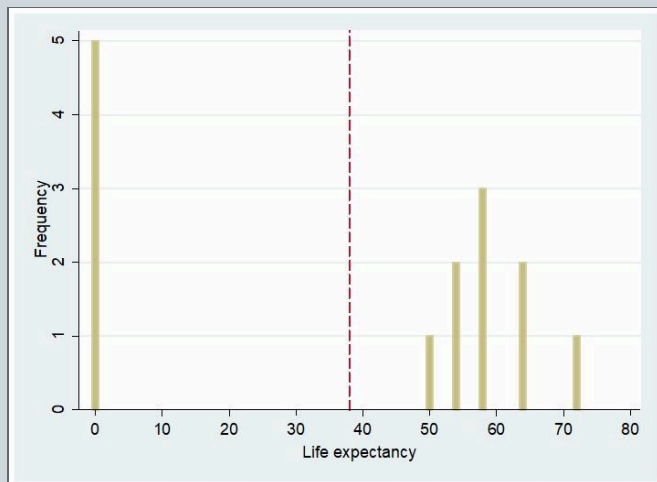
Were people in their late thirties preparing their wills?

No — infant mortality was 30–50%. The average was dragged down by deaths in the first years of life. Survive childhood and you might well see seventy.

The impact of infant mortality



Without infant mortality



With infant mortality

The average does not mean typical.

Central tendency in Python

```
1 pub = news.dropna(subset=["publication_date_one", "publication_date_two"]).copy()
2 pub["length"] = pub["publication_date_two"] - pub["publication_date_one"]
3
4 print("titles with both dates:", len(pub))
5 print("mean  ", round(pub["length"].mean(), 4))
6 print("median", pub["length"].median())
7 print("min   ", pub["length"].min())
8 print("max   ", pub["length"].max())
```

```
titles with both dates: 21764
mean   11.1848
median 3.0
min    0.0
max    348.0
```

Mean (11.18) > median (3): right-skewed. Most newspapers had short lives; a few very long-lived ones pull the mean up.

Spread: quartiles

```
1 print(pub["length"].describe().round(2).to_string())
```

count	21764.00
mean	11.18
std	21.14
min	0.00
25%	1.00
50%	3.00
75%	11.00
max	348.00

- **Min** 0 years
 - **Q1** 1 year — a quarter lasted a year or less
 - **Median** 3 years
 - **Q3** 11 years — three quarters lasted 11 years or less
 - **Max** 348 years
- IQR** = $Q3 - Q1 = 10$ years, the spread of the middle half.

Part IV: Histograms for numeric data

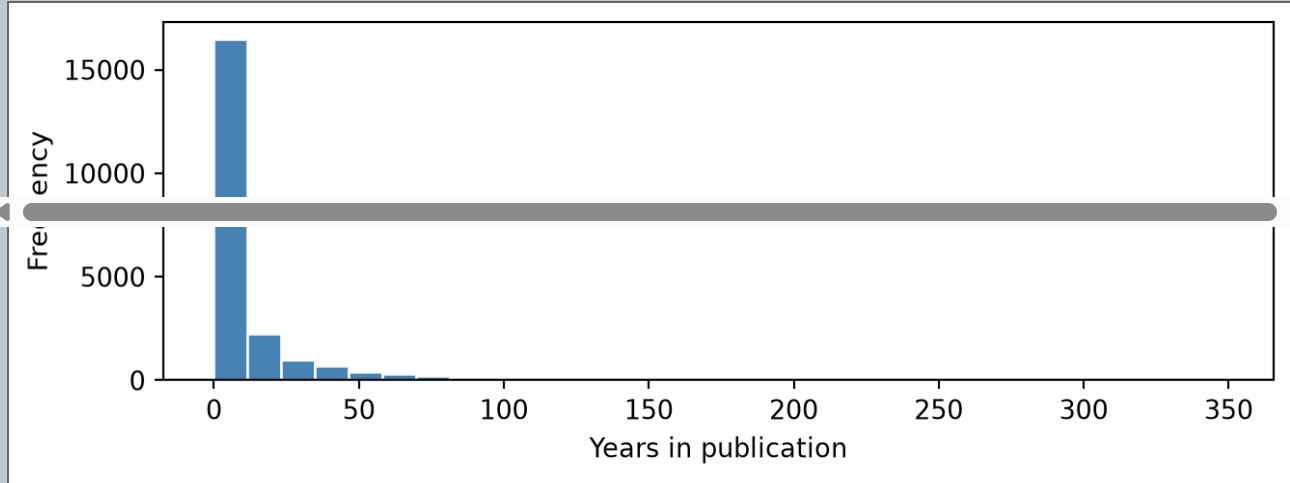
Understanding histograms

- Show the distribution of a **numeric** variable by grouping it into **bins**
- X-axis: value ranges
- Y-axis: count in each bin
- **Bars touch** — unlike a bar plot

Bar plots	Histograms
categorical data	numeric data
gaps between bars	no gaps
order often arbitrary	order meaningful
one bar per category	bins group ranges

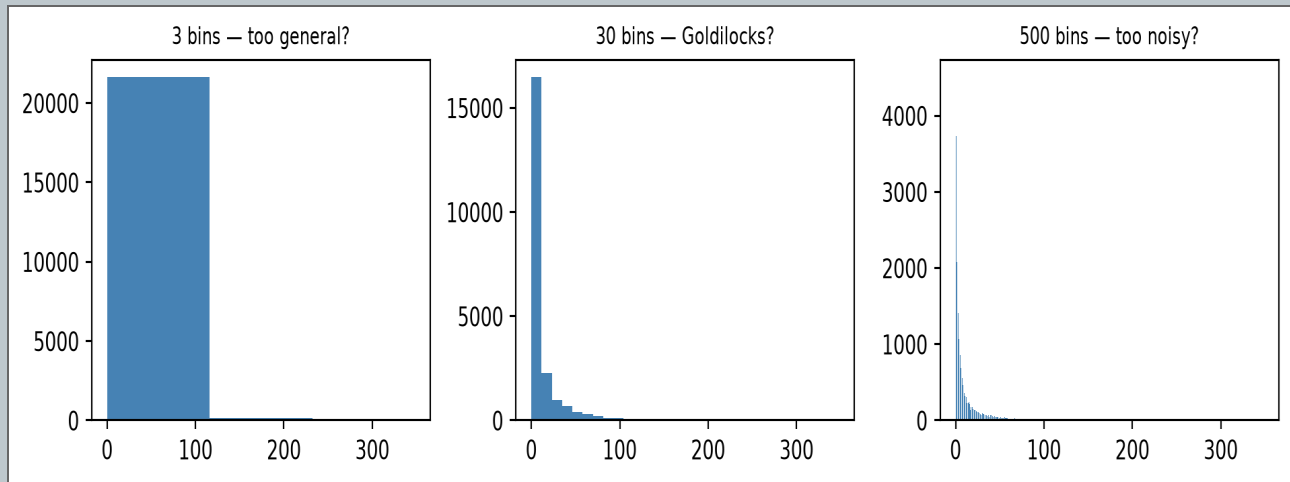
```
1 fig, ax = plt.subplots(figsize=(7, 2.6))
2 pub["length"].plot(kind="hist", ax=ax, bins=30, color="steelblue", edgecolor="white")
3 ax.set_xlabel("Years in publication")
```

```
4 plt.tight_layout()
5 plt.show()
```



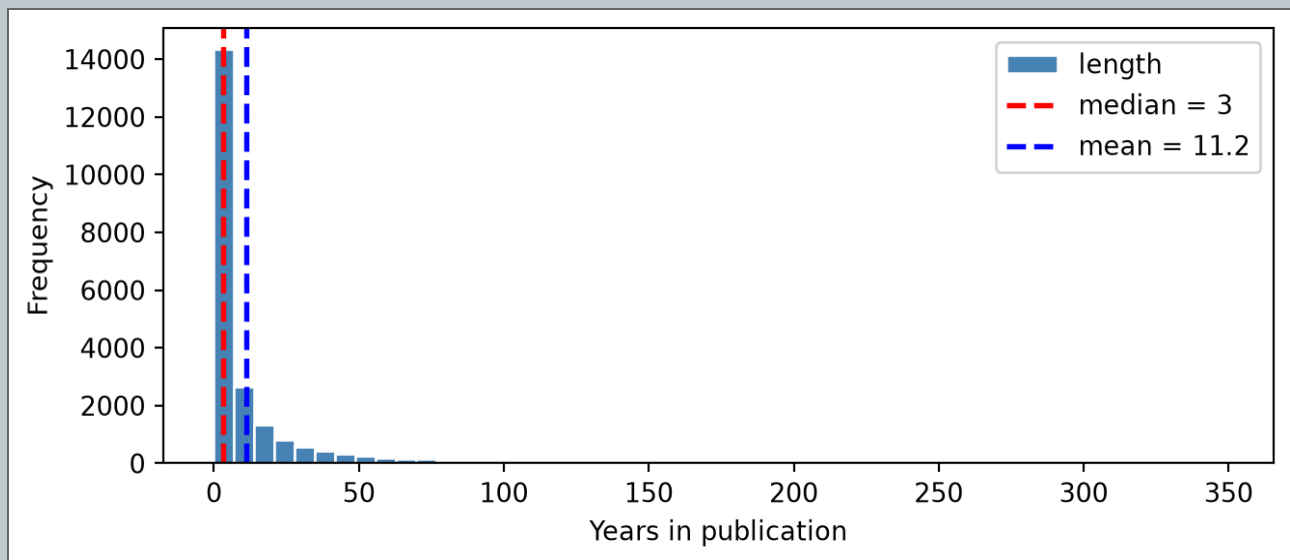
Choosing bins

```
1 fig, axes = plt.subplots(1, 3, figsize=(10, 2.6))
2 for ax, bins, label in zip(axes, [3, 30, 500],
3                               ["3 bins – too general?", "30 bins – Goldilocks?",
4                               "500 bins – too noisy?"]):
5     pub["length"].plot(kind="hist", ax=ax, bins=bins, color="steelblue")
6     ax.set_title(label, fontsize=9)
7     ax.set_ylabel("")
8 plt.tight_layout()
9 plt.show()
```



Adding reference lines

```
1 fig, ax = plt.subplots(figsize=(7, 3))
2 pub["length"].plot(kind="hist", ax=ax, bins=50, color="steelblue", edgecolor="white")
3 ax.axvline(pub["length"].median(), color="red", linestyle="--", linewidth=2,
4             label=f"median = {pub['length'].median():.0f}")
5 ax.axvline(pub["length"].mean(), color="blue", linestyle="--", linewidth=2,
6             label=f"mean = {pub['length'].mean():.1f}")
7 ax.set_xlabel("Years in publication")
8 ax.legend()
9 plt.tight_layout()
10 plt.show()
```



Interpreting a histogram

1. **Centre** — where is the peak?
Mean vs median?
2. **Spread** — how wide? Range and standard deviation
3. **Shape** — symmetric or skewed? One peak or several?
4. **Outliers** — isolated bars?
Gaps?

Our newspaper data

- **Centre:** most titles fail quickly, peak near zero
- **Spread:** enormous, 0–348 years
- **Shape:** extreme right skew
- **Outlier:** the *London Gazette*, 1666–2014

Histogram best practices

Do

- Choose an appropriate bin width
- Label axes clearly, with units
- Consider transformations for skewed data
- Note the sample size
- Add average lines or density curves where useful

Don't

- Use too few bins (hides patterns)
- Use too many bins (creates noise)
- Ignore outliers without investigating
- Compare histograms with different bin widths
- Assume a normal distribution

Part V: Conclusion

Key takeaways

Conceptual

- Visualisation reveals patterns invisible in raw data
- Different data types need different visualisations
- Distribution shape affects interpretation
- Cultural context matters in visualisation

Practice

- Choose the visualisation for the data type
- Label clearly and completely
- Consider your audience
- Save high-quality figures
- Document your choices
- **Think about colour-blind readers**

Resources and references

Documentation

- `help(plt.bar)`,
`help(plt.hist)`
- matplotlib gallery:
<https://matplotlib.org/stable/gallery/>

Books

- Tufte, *The Visual Display of Quantitative Information* (2001)
- Cairo, *The Truthful Art* (2016)

Online

- Data Visualisation Catalogue:
<https://www.datavizcatalogue.com>
- ColorBrewer:
<https://colorbrewer2.org>

Visualisation and cultural analytics

- Du Bois data visualisations:
<https://www.dignityanddebt.org/projects/du-boisian-resources/>
- D'Ignazio & Klein, *Data Feminism*: <https://data->

feminism.mitpress.mit.edu/

Part VI: Live Coding

Live coding

Switch to `w07_live_coding.ipynb`.

1. Bar plot of countries, then fix the labels
2. Crosstab of frequency by country → stacked, grouped, proportional
3. Newspaper lifespans: mean, median, quartiles
4. Histogram, with bin widths compared and reference lines added
5. Save a figure properly

Before next week

1. Complete `w07_worksheet.ipynb` in the workshop
2. **The data overview is due next week** — Monday 16 November
3. Read Hill, “Contextual Analytics”, §5.3

Note

The quiz in next week’s lecture covers **this week’s** material.



Speaker notes